

Practice Final B

MTH 207 (Fall 2026) — J. Baggett

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
 - Explain your reasons and justify your steps.
 - Demonstrate that you understand the *process*.
 - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.

The real final is 120 minutes and 200 points. This practice final is longer (220 points) so that it covers more of the material — do not use it to judge timing.

- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

- (a) If $s(t)$ gives position in feet and t is in seconds, then $s'(3)$ is measured in feet.

Solution:

FALSE. A derivative carries the units of the output divided by the units of the input, so $s'(3)$ is in feet per second.

- (b) If the average rate of change of f on $[1, 3]$ is 0, then $f(1) = f(3)$.

Solution:

TRUE. $\frac{f(3) - f(1)}{2} = 0$ forces $f(3) = f(1)$.

- (c) If f is continuous on (a, b) , then f attains an absolute maximum and an absolute minimum at some numbers in (a, b) .

Solution:

FALSE. The Extreme Value Theorem needs a CLOSED interval. On $(0, 1)$, $f(x) = x$ attains neither.

- (d) If $h(x) = f(g(x))$ with f and g differentiable and $g'(a) = 0$, then $h'(a) = 0$.

Solution:

TRUE. $h'(a) = f'(g(a)) \cdot g'(a)$, and the second factor is zero. Whatever f' contributes is multiplied by 0.

- (e) Substitution can evaluate $\int e^{x^2} dx$ by taking $u = x^2$.

Solution:

FALSE. Then $du = 2x dx$, and there is no x in the integrand to supply it. Substitution needs the inner function's derivative present as a factor.

(7^{pts} ea.) **2.** Evaluate the following limits. As a reminder, justify your steps! Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\lim_{x \rightarrow \infty} \frac{5x + 4}{3x - 7}$

(b) $\lim_{x \rightarrow \infty} (\sqrt{9x^2 + 2x} - 3x)$

(c) $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$

- (10^{pts}) **3.** Sketch the region enclosed by the curves $y = 6 - x^2$ and $y = x$, showing the points where they intersect. Then set up and evaluate an integral for its area.

Solution:

Intersections: $6 - x^2 = x \Rightarrow x^2 + x - 6 = 0 \Rightarrow (x + 3)(x - 2) = 0$, so $x = -3$ and $x = 2$. The points are $(-3, -3)$ and $(2, 2)$.

Which is on top? Test $x = 0$: the parabola gives 6, the line gives 0. So $y = 6 - x^2$ is the upper curve throughout $[-3, 2]$.

$$\begin{aligned} A &= \int_{-3}^2 \left[(6 - x^2) - x \right] dx = \left[6x - \frac{x^3}{3} - \frac{x^2}{2} \right]_{-3}^2 \\ &= \left(12 - \frac{8}{3} - 2 \right) - \left(-18 + 9 - \frac{9}{2} \right) = \frac{22}{3} - \left(-\frac{27}{2} \right) = \frac{125}{6} \approx 20.8 \end{aligned}$$

The sketch is a downward parabola with vertex $(0, 6)$ cut by a line through the origin, meeting it at $(-3, -3)$ and $(2, 2)$.

- (3^{pts}) **4.** Sketch the graph of a function that is continuous everywhere but fails to be differentiable at exactly two x -values. Label those two x -values on your sketch, and say in one sentence what goes wrong at each.

Solution:

$f(x) = |x| + |x - 2|$ works. It is continuous everywhere (sums of continuous functions), with **corners** at $x = 0$ and $x = 2$.

At $x = 0$: the slope jumps from -2 (left) to 0 (right). **At $x = 2$:** the slope jumps from 0 (left) to 2 (right).

At each one the left- and right-hand difference quotients approach different numbers, so the limit defining f' does not exist — a **corner**. The graph is a flat segment of height 2 between $x = 0$ and $x = 2$, with straight arms rising away on either side.

(A vertical tangent, e.g. $\sqrt[3]{x}$, is the other acceptable kind of failure.)

- (20^{pts}) **5.** A streetlight is mounted at the top of a 15-foot pole. A person 6 feet tall walks away from the pole along a straight path at a rate of 5 feet per second. How fast is the tip of the person's shadow moving along the ground when the person is 40 feet from the pole? (Be sure to include units.)

Solution:

Let x be the person's distance from the pole and s the distance from the pole to the **tip of the shadow**, both functions of t .

Similar triangles (pole with the whole shadow; person with the part beyond them):

$$\frac{15}{s} = \frac{6}{s - x}$$

Cross-multiplying: $15(s - x) = 6s \Rightarrow 15s - 15x = 6s \Rightarrow 9s = 15x$, so

$$s = \frac{5}{3}x$$

Differentiate with respect to t :

$$\frac{ds}{dt} = \frac{5}{3} \cdot \frac{dx}{dt} = \frac{5}{3}(5) = \frac{25}{3} \text{ ft/sec} \approx 8.33 \text{ ft/sec}$$

Notice the answer does not depend on how far the person has walked — the tip moves at a constant speed. (The *shadow's length* $s - x$ grows at $\frac{25}{3} - 5 = \frac{10}{3}$ ft/sec, which is a different question.)

(8^{pts} ea.) **6.** Evaluate the following indefinite integrals. Simplify your answer as much as possible.

(a) $\int \frac{x^3 + 2x^2 - x}{x} dx$

(b) $\int \frac{3^{\cos x}}{\csc x} dx$

(c) $\int \frac{\sin(2 + \ln x)}{x} dx$

- (12^{pts}) **7.** The point $(3, 1)$ lies on the curve $x^2y + y^3 = 10$. Find the slope of the curve at that point, and then determine whether the tangent line there passes through the origin. Justify your conclusion.

Solution:

Check the point: $3^2(1) + 1^3 = 9 + 1 = 10$. ✓

Differentiate implicitly (x^2y needs the Product Rule):

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$$

At $(3, 1)$:

$$\frac{dy}{dx} = \frac{-2(3)(1)}{9 + 3} = \frac{-6}{12} = -\frac{1}{2}$$

Tangent line: $y - 1 = -\frac{1}{2}(x - 3)$, i.e. $y = -\frac{1}{2}x + \frac{5}{2}$.

Through the origin? Put $x = 0$: $y = \frac{5}{2} \neq 0$. **No** — the tangent line meets the y -axis at $\left(0, \frac{5}{2}\right)$, not at the origin. (Equivalently, a line through the origin and $(3, 1)$ would have slope $\frac{1}{3}$, not $-\frac{1}{2}$.)

- (10^{pts}) **8.** Given a function $f(x)$, write a formula for the average rate of change of f on the interval $[a, a + h]$. Explain what this quantity measures, and explain how the derivative $f'(a)$ is obtained from it.

Solution:

$$\text{average rate of change} = \frac{f(a + h) - f(a)}{h}$$

What it measures: the change in output divided by the change in input across the interval — geometrically, the **slope of the secant line** through $(a, f(a))$ and $(a + h, f(a + h))$. It is an average over the whole interval, not a value at a point.

How $f'(a)$ comes from it: shrink the interval. Let $h \rightarrow 0$:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

The secant slopes settle on the slope of the **tangent line** at $x = a$ — the *instantaneous* rate of change. h is never 0 in the quotient; it only approaches 0.

(5^{pts} ea.) **9.** Evaluate the following limits. Justify your steps, and name any rule you use.

(a) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$

(b) $\lim_{x \rightarrow 0^+} x \ln x$

10^{pts} ea.) **10.** Evaluate the following definite integrals. Simplify your answer as much as possible.

(a) $\int_0^1 \frac{x}{1+x^2} dx$

(b) $\int_0^{\pi/4} \frac{2+\cos^2 \theta}{\cos^2 \theta} d\theta$

(c) $\int_{-3}^{3\sqrt{3}} \frac{2}{\sqrt{36-x^2}} dx$

- (8^{pts}) **11.** Given a fixed number $a > 0$ with $a \neq 1$, derive the derivative rule for $y = \log_a x$. (You may use the derivative rule for $\ln x$.)

Solution:

Use the change-of-base formula to turn $\log_a$ into $\ln$:

$$y = \log_a x = \frac{\ln x}{\ln a}$$

Here $\ln a$ is a **constant** (it does not involve x), so it just rides along as a constant multiple:

$$\frac{dy}{dx} = \frac{1}{\ln a} \cdot \frac{d}{dx}(\ln x) = \frac{1}{\ln a} \cdot \frac{1}{x}$$

$$\boxed{\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}}$$

Check with $a = e$: $\ln e = 1$, giving $\frac{1}{x}$. ✓

- (15^{pts}) **12.** Use the definition of the definite integral as a limit of Riemann sums to show that $\int_0^3 (4x - 1) dx = 15$. Then confirm your answer using the Fundamental Theorem of Calculus. (You may use any sampling rule, and you may use $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.)

Solution:

By Riemann sums. Take n equal pieces of $[0, 3]$: $\Delta x = \frac{3}{n}$ and (right endpoints) $x_i = \frac{3i}{n}$, so

$$f(x_i) = 4\left(\frac{3i}{n}\right) - 1 = \frac{12i}{n} - 1.$$

$$\int_0^3 (4x - 1) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{12i}{n} - 1 \right) \frac{3}{n} = \lim_{n \rightarrow \infty} \left[\frac{36}{n^2} \sum_{i=1}^n i - \frac{3}{n} \sum_{i=1}^n 1 \right]$$

With $\sum i = \frac{n(n+1)}{2}$ and $\sum 1 = n$,

$$= \lim_{n \rightarrow \infty} \left[\frac{36}{n^2} \cdot \frac{n(n+1)}{2} - 3 \right] = \lim_{n \rightarrow \infty} \left[18 \left(1 + \frac{1}{n} \right) - 3 \right] = 18 - 3 = \mathbf{15}$$

By the Fundamental Theorem. An antiderivative is $F(x) = 2x^2 - x$, so

$$\int_0^3 (4x - 1) dx = F(3) - F(0) = (18 - 3) - 0 = 15 \quad \checkmark$$

(10^{pts}) **13.** Suppose f is continuous on $(-\infty, \infty)$ and satisfies all of the following:

$$f'(x) > 0 \text{ on } (-\infty, 1) \text{ and } (3, \infty), \quad f'(x) < 0 \text{ on } (1, 3),$$

$$f''(x) < 0 \text{ on } (-\infty, 2), \quad f''(x) > 0 \text{ on } (2, \infty).$$

Identify each critical number of f and classify it as a local maximum, a local minimum, or neither, giving your reason. State the x -coordinate of every inflection point. Then sketch a graph that could be f .

Solution:

Critical numbers are where f' changes sign: $x = 1$ and $x = 3$.

- $x = 1$: $f' > 0$ just left, $f' < 0$ just right. By the First Derivative Test this is a **local maximum**.
- $x = 3$: $f' < 0$ just left, $f' > 0$ just right — a **local minimum**.

Inflection point: f'' changes sign only at $x = 2$ (concave down before, concave up after), and f is continuous there, so $x = 2$ is the one inflection point.

Sketch: rising and concave down up to the peak at $x = 1$; falling and still concave down to $x = 2$, where the bending flips; falling but now concave up to the valley at $x = 3$; then rising and concave up thereafter. The inflection at $x = 2$ sits *between* the peak and the valley, on the falling stretch.

(15^{pts}) **14.** Find y' if $y = 2^x \cos x + \ln(\sin^{-1} x) + 3 \int_x^7 \sqrt{1+t^4} dt$.

Solution:

Three independent pieces.

(i) $2^x \cos x$ — Product Rule, with $\frac{d}{dx} 2^x = 2^x \ln 2$:

$$2^x \ln 2 \cdot \cos x + 2^x (-\sin x) = 2^x (\ln 2 \cos x - \sin x)$$

(ii) $\ln(\sin^{-1} x)$ — Chain Rule, with $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$:

$$\frac{1}{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}}$$

(iii) $3 \int_x^7 \sqrt{1+t^4} dt$ — the variable is the *lower* limit, so flip it first: $\int_x^7 = -\int_7^x$. By the Fundamental Theorem,

$$3 \cdot \frac{d}{dx} \left(-\int_7^x \sqrt{1+t^4} dt \right) = -3\sqrt{1+x^4}$$

Adding:

$$y' = 2^x (\ln 2 \cos x - \sin x) + \frac{1}{\sin^{-1} x \sqrt{1-x^2}} - 3\sqrt{1+x^4}$$

- (2^{pts}) **15.** Consider $f(x) = \sqrt{x-1}$ on $[5, 17]$. According to the Mean Value Theorem there is a number c in $(5, 17)$ so that $f'(c) = \underline{\hspace{2cm}}$. Compute and simplify the number to fill in the blank.

Solution:

The Mean Value Theorem says $f'(c)$ equals the average rate of change:

$$f'(c) = \frac{f(17) - f(5)}{17 - 5} = \frac{\sqrt{16} - \sqrt{4}}{12} = \frac{4 - 2}{12} = \frac{1}{6}$$

- (10^{pts}) **16.** Choose ONE integral on this page and evaluate it. Clearly mark which integral is your choice!
(Suggestion: do your scratch work on the back page or ask for extra paper.)

$$\int \frac{e^x}{1 + e^{2x}} dx$$

$$\int \frac{dx}{\sqrt{x}(1+x)}$$

$$\int_0^4 x\sqrt{4 - (x-2)^2} dx$$

$$\int \frac{dx}{\sqrt{-x^2 + 4x}}$$

Solution:

Choosing the first integral. Since $e^{2x} = (e^x)^2$, the denominator is $1 + u^2$ with $u = e^x$.

Let $u = e^x$, so $du = e^x dx$:

$$\int \frac{e^x}{1 + e^{2x}} dx = \int \frac{du}{1 + u^2} = \arctan u + C = \arctan(e^x) + C$$

- (20^{pts}) **17.** A bus company will charge \$200 per person for a tour if 50 people sign up. For each additional person beyond 50 who signs up, the company reduces the price by \$2 per person for everyone on the tour. How many additional people should sign up in order to maximize the company's total revenue?

Solution:

Let n be the number of **additional** people beyond 50.

- people on the tour: $50 + n$
- price each: $200 - 2n$ dollars

Revenue:

$$R(n) = (50 + n)(200 - 2n) = 10000 + 100n - 2n^2$$

$$R'(n) = 100 - 4n = 0 \implies n = \mathbf{25}$$

$R''(n) = -4 < 0$, so this is a maximum.

So **25 additional people** — a tour of 75 people at $\$200 - \$50 = \$150$ each, giving revenue $75 \times 150 = \$11,250$ (against \$10,000 for the bare 50).